Assumptions & Methodology

Standalone companion to the Executive Summary and Detailed Analysis: what was assumed, how each number was produced, what was checked, and what could not be done.

1. Analytical approach in brief

Business question	Method	Evidence strength
1. Who are the highest-risk members?	Transparent point score (cost, hospital use, ED use, chronic burden), four tiers, six segments; prospective test (Jan-Jun score vs Jul-Oct cost)	Strong for describing concentration; weak for prediction
2. Are care-management programs effective?	Enrolled vs not (crude and chronic-only), difference-in-differences with pseudo-dates, engagement dose-response, program funnel	Weak: 68 members with a usable before/after window
3. Where to focus in 2026?	Cost by service, condition, diagnosis, provider, facility; claim denials; UM; ops-metric reconciliation	Strong for service/diagnosis; provider-level not supported

2. Assumptions

Each assumption lists why it was made and what it affects, so you can challenge it.

#	Assumption	Why	Effect / sensitivity
A1	The extract is a ~1,000-member sample (IDs 1-1,000) of ~250,000 lives.	Only 1,000 IDs exist in all files; ever-enrolled care-management share (24.3%) matches the ops enrollment rate (23.7%).	Report rates and relative patterns; never scale dollars to the full book.
A2	Analysis window is 1 Jan-15 Oct 2025 (288 days, ~9.5 months); all members continuously enrolled.	No eligibility file; claims and encounters end 15 Oct.	Per-member-month rates use 9.5 months. Part-year members would make true rates higher.
A3	Cost = plan-paid amount (paid_amount).	Denied claims pay $0; allowed amount would count denied claims as cost.	Excluding PENDING claims would lower totals by 2.6%.
A4	PENDING claims are included at their reported paid amount.	They already carry paid amounts (96% have a paid date).	Small (119 claims, $236k).
A5	Encounter member IDs map to the standard ID by dropping the trailing '01' (integer // 100).	All 3,000 rows share the suffix; 99.2% then match a claims member.	Without this, encounters cannot be joined to any other file.
A6	Inpatient utilization is counted from INPATIENT_ADMIT rows only.	Discharge rows are the same stays (242 admits vs 227 discharges).	Counting both would overstate admissions by ~90%.
A7	A member has a chronic condition if flagged in encounters or coded on any claim (diabetes, cardiac, hypertension, asthma, COPD).	Either source alone under-counts; chronic flags look recorded mainly for care-management members.	Biases comparisons; noted in limitations.
A8	Care-management 'ever' = any enrollment record; 'active' = status ACTIVE; member-level analysis uses first enrollment.	Members can be in several programs.	Counts of enrolled members, not enrollments, in targeting analysis.
A9	enrollment_status (not program_completion_flag) defines completion.	Flag disagrees with status on 66 COMPLETED records.	Completion rate 28% of enrollments by status vs 11% by flag.
A10	Risk is defined by the point score (section 3.3) rather than the program risk_score.	Program score exists only for enrolled members and does not track cost (rho -0.03).	Tiers describe current concentration, not future risk.
A11	readmission_flag is used as provided.	Pre-2025 stays are not visible, so it cannot be recomputed; only 5 of 18 flagged admits show a prior discharge in the extract.	Readmission findings are low-confidence.
A12	Diagnosis codes are taken as recorded.	No way to validate; some look clinically implausible for the setting (e.g. routine exam on inpatient stays).	Condition-level and inpatient-diagnosis findings need clinical validation.
A13	Difference-in-differences comparison group = never-enrolled members with random pseudo-enrollment dates drawn from the treated	Gives controls a comparable before/after split.	Assumes parallel trends, which cannot be tested with ~1-5 months of pre-data.

	dates.		
A14	Claim cost is capped at the 99th percentile (~$31,700) in one version of the cost comparison.	Prevents single large claims dominating small samples.	Uncapped result is also reported; conclusions are the same.
A15	Ops status is re-derived with explicit direction and a +/-5% band; lower MLR is favourable.	The file's labels are direction-aware for most metrics but treat MLR as higher-is-better.	Changes MLR status in several months.

3. Methodology step by step

3.1 Data preparation

- Load all five files; parse dates; check row counts, duplicate keys, nulls and value ranges (full results in the data-quality log below).
- Harmonise member IDs (A5). Set encounter_type to UNKNOWN where missing (2 rows).
- Create paid cost (paid_amount), a P99-capped version, and a diagnosis group from the ICD-10 prefix (Diabetes E1x; Asthma J45; COPD J44; Hypertension I10, I11-I13; Cardiac other I; everything else non-specific/acute).
- Build one row per member (1,000 rows): cost, claim counts, cost by service category, admissions, ED visits, readmissions, preventive visits, chronic conditions, care-management status, UM requests/denials.

3.2 Cost concentration

Members are ranked by total paid cost; the share of total cost held by the top 1%, 5%, 10% and 20% is reported, along with a decile chart.

3.3 Risk score, tiers and segments

Component	Rule	Points
Cost rank (percentile of paid)	>95th / 90-95th / 80-90th	3 / 2 / 1
Admissions	2+ / 1	3 / 1
Readmission	any flagged	+2
ED visits	3+ / 2	2 / 1
Chronic conditions	2+ / 1	2 / 1

Tiers: Low 0-1, Moderate 2-3, High 4-5, Very high 6+ (cut-points chosen so High + Very high is ~7% of members, a realistic outreach capacity; the first cut-points tried (7+ for Very high) left only 7 members). Segments are assigned in order: A hospital-driven high cost (admission and top-decile cost), B frequent ED (3+ visits), C multi-condition chronic (2+), D single chronic, E low-need with primary-care contact, F low-need without.

3.4 Prospective validation

Score rebuilt using only 1 Jan-30 Jun claims and encounters (with within-period cost percentiles); outcomes are paid cost, ED visits and admissions from 1 Jul-15 Oct. Reported with Spearman rank correlation and average later cost by tier. Result: score vs later cost rho = +0.02; first-half vs second-half cost rho = -0.00.

3.5 Care-management targeting

Percent of members ever/currently enrolled within the top 1%, 5%, 10%, 20% and bottom 80% by cost; Spearman correlation of enrollment risk_score with cost and ED visits among enrolled members.

3.6 Care-management effectiveness

- **Crude and chronic-only comparison:** cost, ED visits and admissions per member-month for enrolled vs not enrolled (and vs not-enrolled members with a chronic condition). ED rate ratios use a log-normal approximation for 95% intervals.
- **Difference-in-differences:** treated = 68 members whose first-ever enrollment was 1 Feb-31 Jul 2025; comparison = 757 never-enrolled members with random pseudo-dates drawn from the treated dates. For each member the monthly rate before and after the date is computed (cumulative daily matrices); estimate = (treated after - before) - (comparison after - before). Point estimate is the mean of 200 random pseudo-date draws; 95% intervals are the 2.5th-97.5th percentiles of 600 bootstrap resamples (members and dates resampled). Outcomes: capped cost, uncapped cost, ED visits, admissions.
- **Robustness:** removing the 5 treated members with the largest cost increase changes the cost estimate from +709 to +178 per member-month. Four outcomes were tested, so a single borderline result is not treated as significant.
- **Dose-response:** among treated members, change in cost and ED rate by care plan/assessment, contact rate >= 50%, goal attainment >= 50%.
- **Operations:** engagement funnel, program and care-manager profiles, tenure of active enrollments, repeat and overlapping enrollments.

3.7 Focus areas

- **Spend drivers:** paid cost by service category and diagnosis group; condition cohorts (members can overlap) with cost per member relative to other members.
- **Inpatient diagnosis review:** inpatient claims coded to non-specific/routine diagnoses; a 'low-acuity' pool is Z00.00, J06.9 and R51.
- **Provider variation:** observed/expected paid cost per provider (expected = average paid for the service category of each claim); spread compared with 300 permutations that shuffle paid amounts within service category. Provider denial rates tested with a one-sided binomial test ($p<0.01$).
- **Facilities:** admissions, length of stay and inpatient cost by facility; not ranked because of small counts.
- **Denials and UM:** claim denials by reason; UM denial, appeal and overturn rates (overturned / (overturned + upheld)); turnaround = decision date - request date.
- **Trend:** paid cost per member-month by incurred month, adjusted to a 30.4-day basis; September and October flagged as incomplete.
- **Ops reconciliation:** metrics re-derived (A15) and recomputed from detail files where possible.

3.8 Parameters and reproducibility

Parameter	Value
Random seed	2025 (fixed)
Pseudo-date draws / bootstrap resamples	200 / 600
Cost cap	99th percentile of claim paid ($31,694)
Permutation tests for provider spread	300
Ops status band	+/-5% of target
Observation window	288 days (1 Jan-15 Oct 2025)
Software	Python 3 (pandas, numpy, scipy, matplotlib, python-docx)

Run scripts/analysis.py then scripts/build_reports.py; every number in the reports comes from outputs/results.json and outputs/tables/.

4. Data-quality findings and handling

Area	Issue	Records	How handled
encounters	member_id is 14 chars (MBR-0000097101) vs 12 chars elsewhere; trailing '01' suffix on every row	3,000	Stripped suffix (integer // 100) -> 99.2% of encounter rows then match a member in claims
all	Extract has member IDs 1-1000 only (994 with claims) vs ~250,000 covered lives in the scenario	1,000	Treated as a ~1,000-member sample. Report relative patterns/rates, never extrapolate $ totals to 250k
all	No members/eligibility file: no demographics, enrollment dates, premiums or member-months	0	Assumed all 1,000 members continuously enrolled for the 288-day window (Jan 1 - Oct 15, 2025)
claims	Claims window ends 2025-10-15 (not full-year 2025); paid dates run to 2025-11-28 (after the scenario's Oct 17 date)	427	Used incurred (claim_date) for trend; flagged Sep-Oct as incomplete; kept paid_amount as given
claims	DENIED claims carry a paid_date (dictionary: null for denied) and member_responsibility > 0 ($172,693 total)	284	Paid amount is $0 on all denied claims, so cost = paid_amount; member_responsibility on denials noted but excluded from plan cost
claims	PENDING claims already have paid_amount > 0 (and 96% a paid_date)	119	Included in plan cost (paid_amount as reported); sensitivity: excluding them changes totals by 2.6%
claims	Diagnosis codes look weakly related to setting: e.g. Z00.00 (routine exam), J06.9 (URI), R51 (headache) appear as the diagnosis on inpatient stays costing $20k+	192	Kept as recorded; flagged as a coding / clinical-validation finding (or synthetic-data artefact). Condition analysis uses both claim dx and encounter flags
claims	310 inpatient claims vs 242 inpatient-admit encounters for the same members	68	Counted utilisation from encounters (admits) and cost from claims; not reconciled 1:1 (no shared key)
encounters	encounter_type missing	2	Left as 'UNKNOWN'; 1 has an inpatient-style stay, 1 has COPD flag - excluded from type-specific counts
encounters	Each inpatient stay appears as an ADMIT row and (for most) a DISCHARGE row (242 admits vs 227 discharges)	469	Counted stays from INPATIENT_ADMIT rows only to avoid double counting
encounters	readmission_flag TRUE on 18 admits but only 5 admits fall within 30 days of that member's prior discharge in this extract	18	Used the provided flag for reporting (cannot see pre-2025 stays) and noted that it cannot be reproduced; recommend validating the flag logic
care_mgmt	risk_score above documented 0-10 range	5	Kept; capped at 10 when used in comparisons
care_mgmt	enrollment_status=COMPLETED but program_completion_flag=FALSE	66	Used enrollment_status as the completion indicator; flag reported separately
care_mgmt	disenrollment dates later than the data window / scenario date (up to 2026-04-11) on COMPLETED/DISENROLLED rows	43	Treated as scheduled/future-dated: enrollment considered open through the window for those rows
care_mgmt	Same member enrolled in the same program more than once (repeat enrollments)	271	Kept (re-enrollment is plausible); member-level analyses use first enrollment
care_mgmt	209 of 400 enrollments began in 2024 but claims start Jan 2025 - no pre-enrollment claims for those members	209	Pre/post analysis restricted to members first enrolled Feb-Jul 2025
um	service_type vs procedure_code mismatches (e.g. DME and Home health coded with office-visit E/M codes 9921x)	442	Not used analytically; flagged
ops	performance_status is direction-aware (+/-5% band) for most metrics but Medical Loss Ratio is labelled as if higher=better (e.g. 96.6% vs 85% target = ABOVE_TARGET)	10	Re-derived favourable/unfavourable status with explicit direction (lower MLR = favourable to plan margin)
ops	Free-text notes are inconsistent with values (e.g. 'Data quality issue resolved' on Member Satisfaction 4.32; 'Below target - action plan needed' once)	9	Ignored notes

5. Checks performed

- Independent recomputation of cost concentration, enrollment shares and member counts in separate exploratory code before the final script (results matched).
- Joins verified: encounter IDs after harmonisation (99.2% match), care-management and UM members present in claims (98% and 99.8%).
- Sensitivity of cost results to PENDING claims, claim capping and removal of the largest changes (section 3.6).

- Provider and denial 'outliers' tested against chance rather than ranked.
- Full re-run from raw files produced identical results (fixed seed).

6. Limitations

- No eligibility, demographics, premium or budget data: no member-months, PMPM, MLR or age/sex adjustment; the 'cost above budget' premise is untested.
- Short window and 2025-only claims: no baseline for the 209 enrollments that began in 2024; effectiveness rests on 68 members.
- Chronic flags appear to be recorded mainly for care-management members, and diagnosis codes look weakly tied to setting.
- Utilization in the extract is far above typical commercial levels, so the sample may over-represent heavy users.
- Recent months are incomplete because claims arrive ~27 days after service on average.
- Observational design: no randomisation, so differences cannot be attributed to the programs.